

Casio FZ-1 Memory Expansion

Assembly Manual

Memory Expansion is designed to add an extra 1MB of sample memory for Casio FZ-1. It replicates the 1987 circuitry by leveraging off the shelf 256Kx1 DRAM chips which are easily obtainable.

The kit does not include DRAM memory chips. These must be sourced separately by the user. While we do not stock vintage DRAM ICs, compatible chips are inexpensive and readily available online.

BOM:

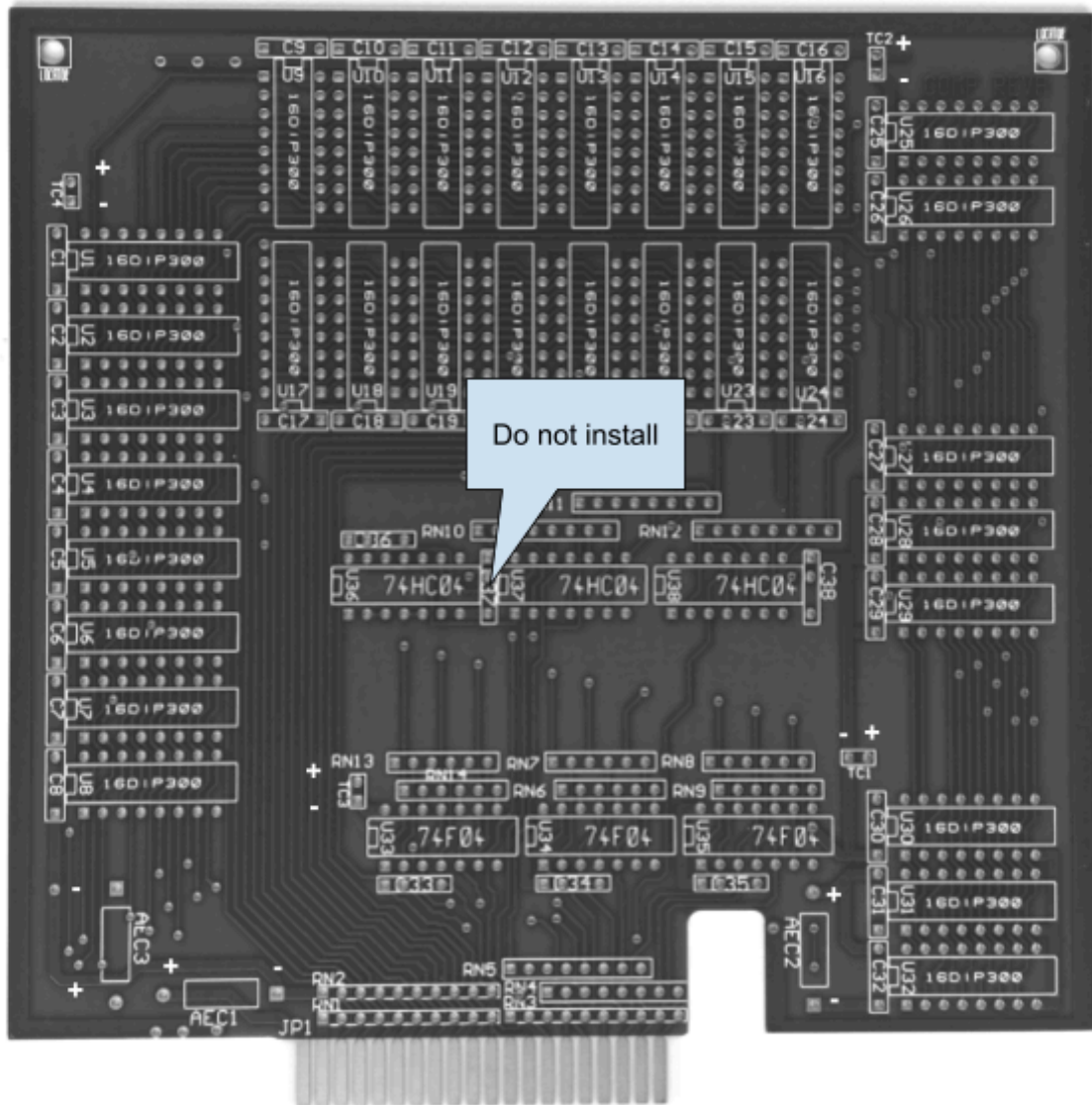
	Location	P/N
IC		
	middle	595-SN74HC04AN
	bottom	595-SN74F04N
	all others	53c256-80
Capacitors		
	TC1-TC4	UST1E100MDD1TE
	C01-C36	80-C420C104M5U5HA
	C37	80-C320C330K2G
	AEC1-AEC3	594-2222-021-36101
Resistors		
	RN1, RN2	71-CSC10A014K70GEK
	RN3	652-4610X-2LF-100
	RN4, RN5	652-4608X-2LF-100
	RN6-RN9, RN13, RN14	652-4606X-2LF-33
	RN10-RN12	858-L083S330LF

The expansion is compatible with any 256Kx1 DRAM rated at 80 ns or faster, including:

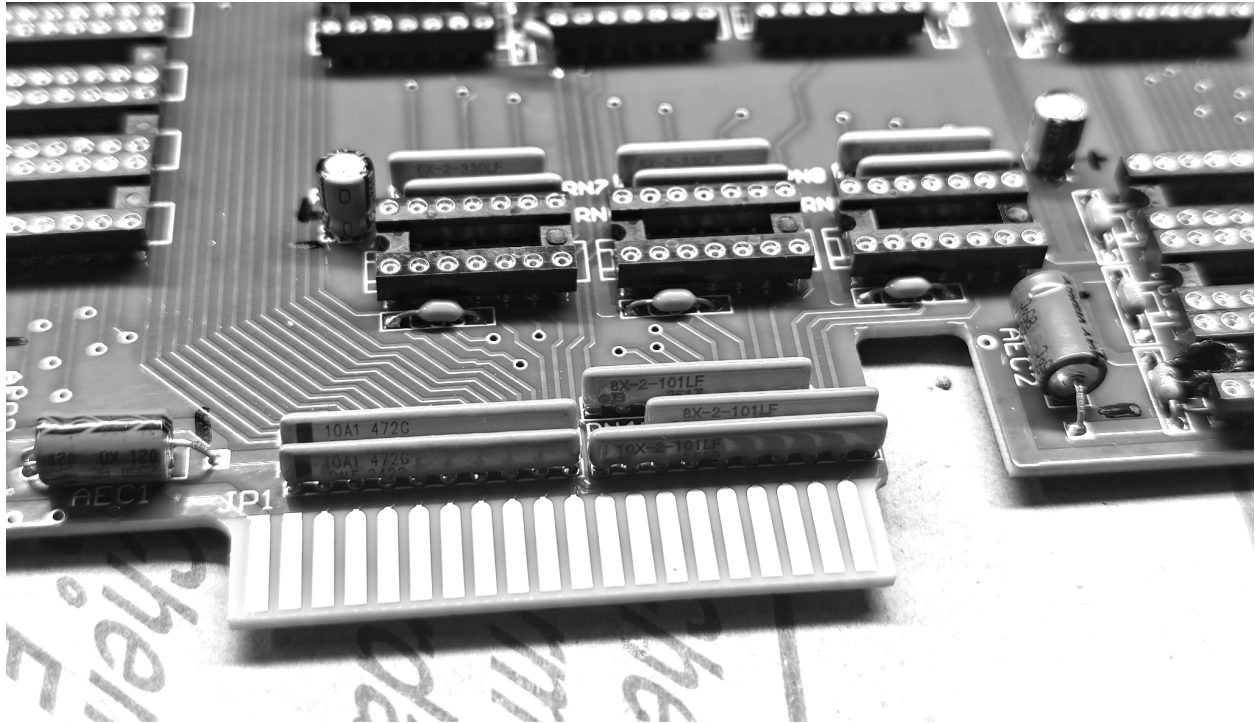
- KM41C256P-7
 - MB81256-80
 - HY53C256LS-80
 - MN41256A-08
- ...and many equivalent parts.

Assembly steps:

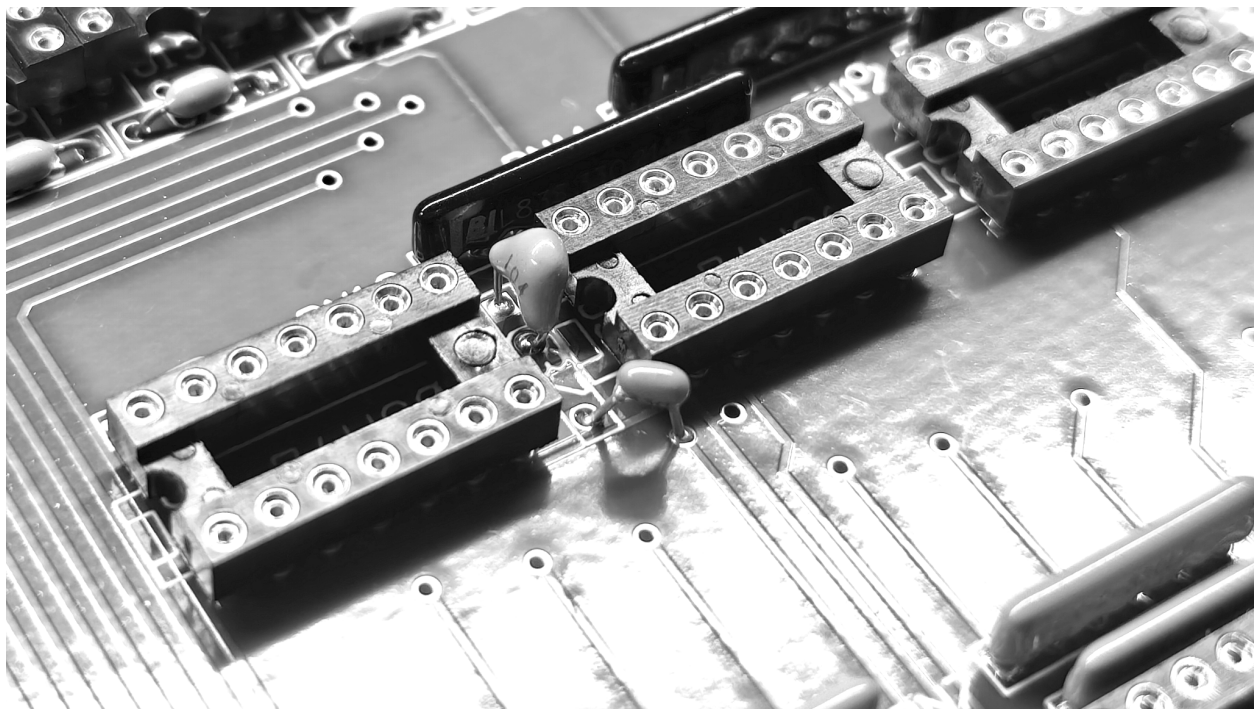
1. Solder all decoupling capacitors first. They are nonpolarized and are located next ICs. Do not solder at C37 place. This one is special.



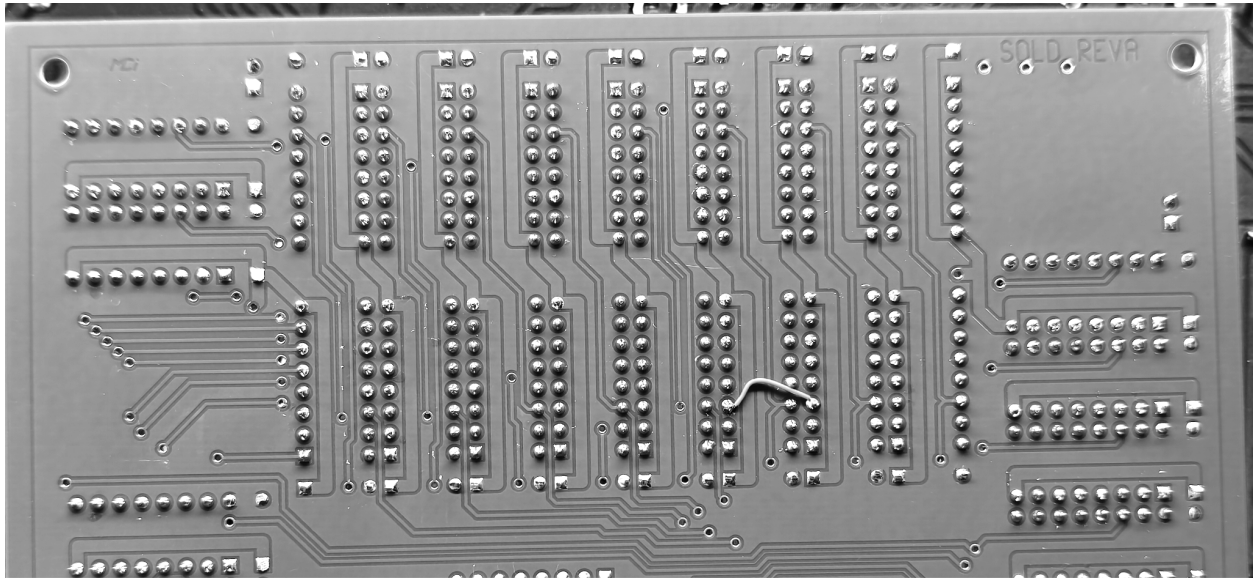
2. Solder all IC sockets
3. Solder all resistor networks, follow pin1 orientation - hint, all text label should face to the edge connector
4. Solder remaining capacitors, look below for the polarity.
 - a. Square pad is typically a negative terminal.
 - b. On large 100uF axial caps a band goes to the positive terminal



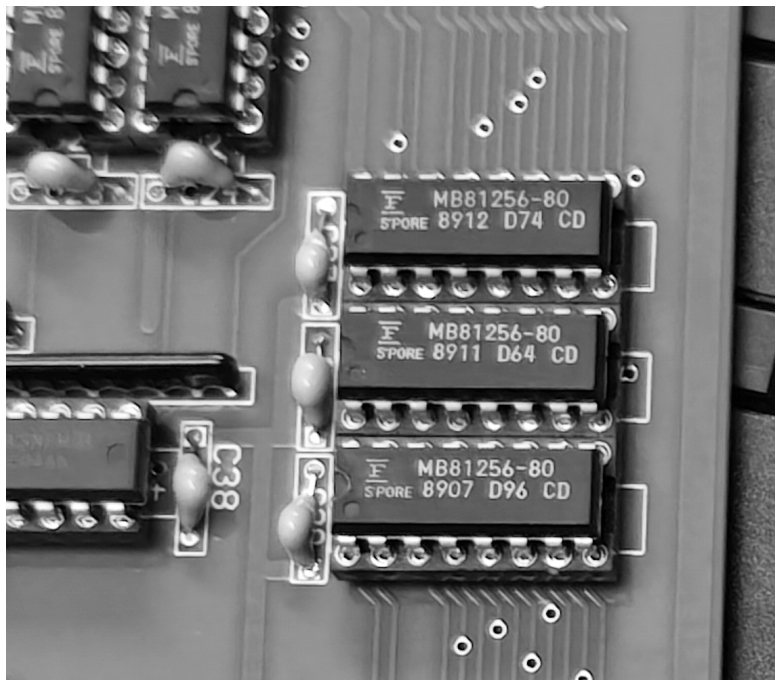
5. Solder C37 as shown
 - a. 104 cap goes to two pins between IC sockets
 - b. While 33pF connects to pins on the corner



6. Add a wire on the back side. This is a factory rework, the replica board follows it as well.
 - a. A wire connects Pin11 of U18 and U19



7. Populate U36, U37, U38 with 74HC04
8. Populate U33, U34, U35 with 74F04
9. Install memory chips. Please follow the silkscreen markings carefully and ensure correct IC notch orientation during installation.



Important Handling Notice:

DRAM memory chips are highly sensitive to static electricity and can be permanently damaged by improper handling. Always use appropriate anti-static precautions when installing or handling memory ICs, including grounding yourself before touching components and avoiding contact with chip pins whenever possible. It is strongly recommended to work on an anti-static surface and use an ESD wrist strap during assembly.

To install the board:

1. Insert the expansion board into the memory expansion slot with all components facing upward.
2. The board should slide smoothly along the rails, requiring only slight pressure at the end for proper seating.
3. Before installation, it is recommended to inspect and clean the expansion slot if necessary. Dust or oxidized contacts may prevent reliable operation. Use DeoxIT or another suitable contact cleaner if needed.

The expansion is fully plug-and-play. On the next power-up, the Casio FZ-1 should automatically detect the additional memory.

For verification, it is recommended to boot the “hardware_checker” utility and perform a memory test to confirm that the expansion is properly recognized and operating without errors.

You can download the floppy emulator image here:

[https://github.com/Deftaudio/Casio-FZ-1/tree/main/Floppy%20Emulator%20Files%20\(FlashFloppy\)/](https://github.com/Deftaudio/Casio-FZ-1/tree/main/Floppy%20Emulator%20Files%20(FlashFloppy)/)

